

Unit: Unit 6: Systems of Linear Equations and Inequalities

Topic: Applications of Systems of Inequalities

Name: _____

Date: _____

Advanced (Habanero)

Open Ended Questions (FRQ)

Q9. A factory produces two types of products, A and B. Each unit of product A requires 2 hours of labor and each unit of product B requires 4 hours of labor. The factory has 240 hours of labor available per week. If the profit from each unit of product A is 10 and the profit from each unit of product B is 15, find the maximum profit the factory can make per week.

Q10. A farmer has 15 acres of land to plant wheat and barley. Each acre of wheat requires $\frac{1}{2}$ ton of fertilizer and each acre of barley requires $\frac{3}{4}$ ton of fertilizer. The farmer has 10 tons of fertilizer available. If the profit from each acre of wheat is 50 and the profit from each acre of barley is 75, find the maximum profit the farmer can make.

Chef's Tips

- Read the questions carefully
- Use a pencil to answer the questions
- Check your work before submitting